MIRROR

Multimodal Intelligent Radiology Reasoning and Observation Reporter

A three-layer radiology pipeline whose language layer reads structured evidence rather than pixels; three modality taxonomies are registered and routed, one of which, chest X-ray, is trained and benchmarked here.

Vignesh Nagarajan
Corresponding Author
Brown Foundation Scholar, Texas A&M University
Department of Computer Science & Engineering
vigneshn26@tamu.edu

Sriram Venkatapathy
Research Mentor
Applied AI Researcher, Capital One
PhD in Computer Science, IIT Hyderabad
sriram.venkatapathy@gmail.com

Global Indian Scientists & Technocrats (GIST) 2026 Summer Research Internship

Abstract. Most medical-imaging models stop at a prediction, offering neither the location of the evidence nor an account a clinician can read. We present **MIRROR**, a three-layer radiology pipeline composing a multi-label classifier, class-activation evidence localization, and a report generator that receives only structured upstream evidence: a per-label probability, a present/absent flag, and a saliency-derived anatomical region. Because the language layer never receives the image, the report cannot assert a finding the classifier did not produce. That guarantee is *finding-level, not detail-level*: the descriptive prose framing each finding is generated text, is not checked against the image, and we give a worked example in which it is measurably ungrounded. One registry supplies the taxonomy, anatomical vocabulary, and report phrasing for chest X-ray (14 findings), brain MRI (11), and head CT (11), with routing explicit or read from the DICOM modality tag; all three are exercised by the test suite, and one, chest X-ray, is trained and benchmarked. On ChestMNIST, a downsampled ChestX-ray14 derivative with the identical 14-label taxonomy, a DenseNet-121 reaches macro AUROC 0.729 (95% CI $[0.718, 0.738]$) with a per-label ordering that tracks finding visibility. At the default 0.5 threshold that same model emits no positive prediction whatsoever for 11 of the 14 labels, and macro F1 is 0.031. Its calibration nonetheless looks excellent (Brier 0.045, ECE 0.018), yet a constant predictor that ignores the image and always returns each label's base rate scores macro Brier 0.047 on the same test set. We report this as a finding: under heavy multi-label class imbalance, aggregate calibration metrics are close to uninformative, and a model silent on most of its taxonomy still posts excellent-looking ones. All headline numbers carry bootstrap 95% confidence intervals and every run stamps a reproducibility record.

1 Introduction

Deep networks now match specialists on narrow chest-radiograph reading tasks [2], yet clinical adoption remains limited by a trust gap: a bare probability tells a radiologist neither *where* the model is looking nor *why* it concluded what it did, and high-stakes decisions demand models that can be interrogated [12]. Saliency methods such as Grad-CAM [8] expose *where* a prediction comes from, and language models can render structured evidence as prose, but the two are rarely composed into one system whose outputs stay mutually consistent.

MIRROR composes them: image $\rightarrow$ prediction $\rightarrow$ localization $\rightarrow$ reasoning $\rightarrow$ report. Its central constraint is *grounding*: the language layer receives only the structured evidence produced upstream (labels, probabilities, named saliency regions), never the raw image, so the report can assert only findings the classifier predicted and the localizer situated. Only the finding taxonomy, the anatomical vocabulary, and a little report phrasing differ between modalities, so these are centralized in one registry and a study is routed to the correct classifier head explicitly or from its DICOM modality tag.

Research question. Can a radiology pipeline be built so that its language layer is structurally incapable of asserting a finding the classifier did not detect, and what does that constraint cost? We answer the first half architecturally and the second empirically, on chest radiographs. Two things this paper does *not* answer should be stated at the outset. It does not measure explanation *quality*: our harness defines a pointing-game and IoU protocol against the NIH lesion boxes, but running it needs a checkpoint trained on the full-resolution release, so no box-level score is reported here and none should be inferred. Nor does it measure whether any of this helps a human, which would need a reader study with radiologists timed and scored on the same studies with and without the explanation and report layers.

Contributions.

1. The distinction between *finding-level* grounding, which constraining the language layer to structured evidence does buy and which is checkable, and *detail-level* ground-

ing, which it does not buy and which we show failing in a real generated report.

2. A registry-driven multimodality in which a modality's taxonomy, anatomical vocabulary, and report phrasing are data rather than code. Three taxonomies are registered, routed, and exercised by the test suite; one is benchmarked.
3. An open-source implementation: three backbones, two explainability methods, an LLM report backend with a deterministic offline fallback, a real DICOM ingest, and two very different inference engines behind one response contract (Fig. 2).
4. Two negative results the harness surfaces on our own model: silence across most of the label set at the default threshold, and aggregate calibration metrics that barely separate a trained classifier from a constant.

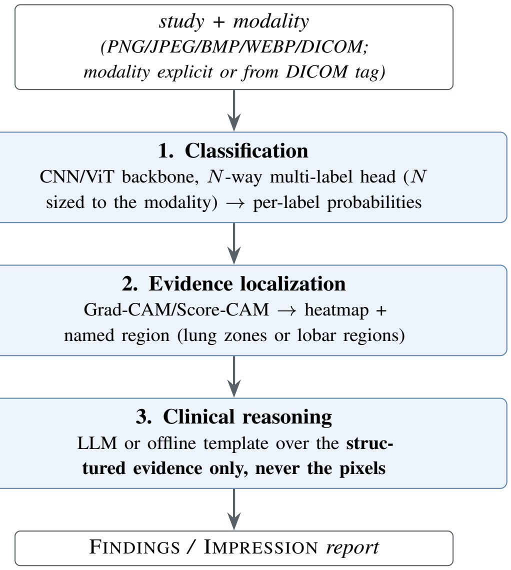


Figure 1: The MIRROR pipeline as deployed in the local PyTorch stack, which is the configuration benchmarked in Section 5. Layers 2 and 3 are individually toggleable, which recovers the conditions of Section 4. The hosted engine of Section 3.5 does not have this structure.

2 Literature Review

Classification and visual explanation. The NIH ChestX-ray8/14 release [1] (112,120 frontal radiographs, 14 pathologies) made supervised deep learning on this modality practical, and MIMIC-CXR [3] added paired free-text reports. CheXNet [2] showed a 121-layer DenseNet [5] reaching radiologist-level pneumonia detection; EfficientNet [6] and Vision Transformers [7] complete the standard toolkit, and MIRROR keeps all three interchangeable with DenseNet-121 primary. On the explanation side, Grad-CAM [8] localizes the evidence behind a CNN prediction via gradients and Score-CAM [9] removes the gradient dependence. The pointing-game protocol [10] and IoU against annotated lesions turn heatmaps into quantitative scores; our harness implements both, though we do not run them here.

Critiques of post-hoc explanation. Adebayo et al. [11] showed several saliency methods can be insensitive to the model and the data they purport to explain, and argued for sanity checks before any map is trusted. Rudin [12] argued more sharply that high-stakes decisions call for inherently interpretable models rather than post-hoc explanations of black boxes. MIRROR is squarely the kind of system these critiques target, and we take up its position against them in Section 6 rather than citing them as background.

Report generation. Systems emitting radiology prose range from TieNet [13] to memory-driven transformers such as R2Gen [14] to decoders optimized for factual correctness [15]. The recurring failure mode is hallucination: prose generated directly from pixels can assert findings the model never detected. Surveys of medical XAI [16] note that interpretability and report generation are usually studied separately. MIRROR's contribution is neither a new classifier nor a new saliency method but their composition under a constraint that bounds what the report may assert.

3 System Architecture

MIRROR chains three layers; each layer's output is the grounded input to the next (Fig. 1).

3.1 Layer 1: Classification

The classifier is a factory over three ImageNet-pretrained backbones: DenseNet-121 (default), EfficientNet-B0, and ViT-B/16, with a multi-label head whose width N is the size of the active modality's taxonomy (14 for chest X-ray, 11 each for brain MRI and head CT; sigmoid applied at loss and inference time, so the network emits raw logits). Inputs are resized to the backbone's 224×224 input and normalized with ImageNet statistics. Training uses `BCEWithLogitsLoss`, AdamW, a cosine schedule, and dropout 0.2, selecting on best validation macro AUROC.

DICOM ingest. Real radiology data arrives as DICOM, and the gap between that and a PNG is not a file-format detail. Stored pixel values are not display values: they must pass through the modality LUT (rescale slope and intercept, which is what turns raw CT counts into Hounsfield units) and then the VOI or window LUT that selects the diagnostic contrast range, and a `MONOCHROME1` photometric interpretation means the stored values are inverted relative to the usual convention. Skipping any of these silently trains a model on images no radiologist would recognize, most visibly on `MONOCHROME1` studies, which arrive as photographic negatives of what they should be. MIRROR ships an ingest that applies all three before the pipeline sees a pixel, lifts only non-PHI technical tags, and exposes the modality tag for routing.

3.2 Layer 2: Evidence Localization

For each positive label (top-k, k=3 by default, to bound compute), this layer computes a class-activation map, either Grad-CAM (forward/backward hooks on a per-backbone target layer) or Score-CAM (gradient-free); ViT maps are recovered by reshaping the patch-token sequence back to a 14×14 grid. The activated region's centroid is mapped into a 3×3 anatomical grid in radiology convention (patient-right on the viewer's left), with the modality's plane picking the vocabulary: lung zones for a frontal chest film, lobar regions for an axial brain-MRI or head-CT slice.

3.3 Layer 3: Clinical Reasoning

This layer converts structured evidence (labels, probabilities, named regions) into a FINDINGS/IMPRESSION report via a system prompt plus an evidence-grounded user prompt with per-label plain-English glosses. The backend is an LLM (Claude) at low temperature, with a deterministic offline template fallback on any error, so the system yields a coherent report with no network or API key. Predictions below the confidence threshold (0.5) surface as pertinent negatives rather than being dropped. Because both backends consume identical evidence, swapping them changes *how fluently* the report reads, not which findings it asserts; and because the layer never receives the image, it cannot introduce findings absent from the upstream evidence. The grounding is therefore at the level of findings, not of the prose framing them, a distinction that carries most of this paper's argument (Section 6).

3.4 Multi-modality via a modality registry

Swapping a chest radiograph for a brain MRI or head CT changes only the taxonomy the head predicts, the vocabulary a saliency centroid is named in, and a little report phrasing (so a normal brain study reads "no acute intracranial abnormality," not "no acute cardiopulmonary abnormality"). These live in one registry (Table 1), the single source of truth for each modality's labels, glosses, plane, and report guidance; a hand-kept TypeScript mirror gives the hosted engine and UI the identical taxonomy and *ordering* (output index i maps to labels$[i]$). Modality is resolved explicitly or, for DICOM, from the (0008,0060) tag, and the orchestrator caches one engine per modality on demand. The claim is architectural and scoped as such: routing, head sizing, vocabulary, and phrasing are exercised by the test suite for all three taxonomies, but no brain-MRI or head-CT checkpoint has been trained, so MIRROR makes no predictive claim on those modalities.

3.5 Orchestration and serving

A single `analyze()` entry point runs the stages, records per-stage timings, and returns one result object; layers 2 and 3 are individually toggleable. Two engines satisfy the same JSON contract (Table 2), so the web UI (Fig. 2) is identical either way: a finding carries *either* a rendered Grad-CAM overlay (local) *or* a normalized bounding box (hosted).

Table 1: The modality registry. Registration means routing, head sizing, and anatomical vocabulary are implemented and tested. Only chest X-ray is trained and benchmarked here.

Modality	#lbl	Taxonomy source	Location vocab.
Chest X-ray	14	NIH ChestX-ray14	lung zones (frontal)
Brain MRI	11	Brain Tumor MRI + neuro findings	lobar regions (axial)
Head CT	11	RSNA ICH subtypes + acute findings	lobar regions (axial)

Table 2: Two inference engines behind one response contract. Only the local stack implements the grounded architecture of Fig. 1, and only it is benchmarked here.

	Local stack	Hosted (serverless)
Engine	FastAPI + PyTorch	Next.js route, vision LLM
Classify	DenseNet/EffNet/ViT	LLM scores N labels
Localize	Grad-CAM/Score-CAM PNG	LLM returns a bbox
Report	LLM or template	LLM, same call
Sees pixels?	Layer 1 only	Every stage
Inputs	+DICOM, BMP	PNG/JPEG/WEBP

The two engines are not two implementations of one design, and the difference matters for every claim here. The local stack is the architecture of Fig. 1. The hosted engine exists because that stack does not fit serverless limits, and replaces the whole thing with a single vision LLM under a forced tool call returning the per-label probabilities, one box per present finding, and the report in one response. That engine reads pixels directly, so MIRROR's grounding constraint **does not hold on the hosted path**: nothing structurally prevents its prose from asserting a finding its own probability vector does not support. Everything benchmarked in Section 5 is the local stack, and hosted-engine figures are labelled as such.

4 Experimental Setup

Data. MIRROR targets NIH ChestX-ray14 [1] (112,120 radiographs, 14 multi-label pathologies). Because the full release is 45 GB and needs a GPU, we evaluate on **ChestMNIST** [4], its downsampled derivative in MedMNIST v2 (CC BY 4.0), which carries the *identical* 14-label taxonomy in the identical order, so MIRROR runs on it unchanged. We sample 8,000 studies from the pooled official train and validation splits and hold out 10%, giving 7,200 training and 800 validation images, and evaluate on 12,000 held-out test images. Training is DenseNet-121 on CPU, AdamW at learning rate 3×10^{-4}, weight decay 10^{-5}, batch 32, best of a four-epoch run on validation macro AUROC. Source images

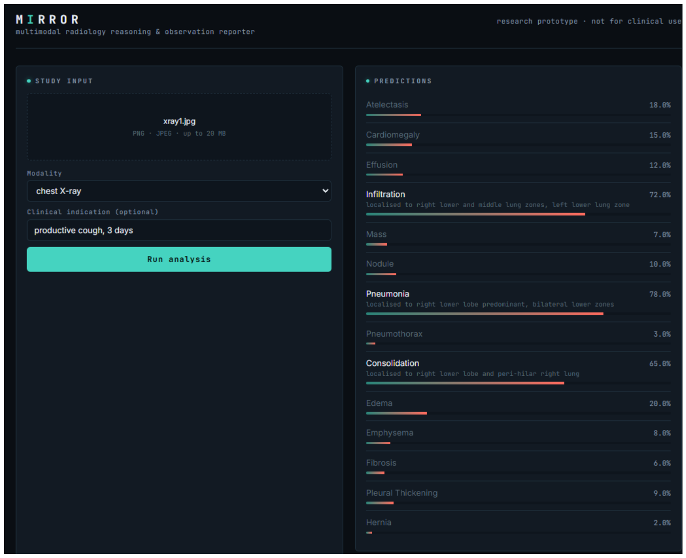


Figure 2: The MIRROR "reading-room" web UI, showing a session on the **hosted serverless engine**, where a vision LLM reads the image directly and returns probabilities, boxes, and report text. This is *not* the benchmarked DenseNet-121 local stack, and no number in this figure is a measured result: it illustrates the interface and the response contract only. A chest radiograph is uploaded with the indication *"productive cough, 3 days,"* all 14 ChestX-ray14 labels are scored, and findings above threshold surface with a location. Both engines drive this identical interface.

are 64×64 and are *upsampled* to the backbone's 224×224 input: there is no genuine 224-pixel information anywhere in this experiment, and findings whose radiographic signature is a few millimetres wide are largely destroyed before training begins.

Metrics. *Discrimination:* per-label and macro AUROC, and macro AUPRC. *Operating point:* sensitivity, specificity, PPV, and NPV at the 0.5 threshold, the quantities a clinician reads off a model. *Calibration:* Brier score and Expected Calibration Error (10 equal-width bins). Each headline metric carries a bootstrap 95% confidence interval: the test set is resampled with replacement $B{=}1000$ times, a *single* resample driving all metrics so the intervals are mutually consistent, reported as two-sided percentile intervals ($\alpha{=}0.05$).

Post-hoc regression test. Three conditions, `classification_only`, `with_localization`, and `full_mirror`, run the same studies with layers 2 and 3 switched on and off. Those layers modify no weights and no logits, so the predictions must be identical across conditions; the harness verifies this and profiles the per-stage wall-clock cost of each added layer. This is a test that the implementation matches the design, not an experiment with an uncertain outcome, and Section 5.5 reports it as such. Every results file stamps a reproducibility record (seed, git commit, library versions).

5 Results

Every number below is measured on this machine with the local PyTorch stack and traces to a JSON committed in the repository.

5.1 Discrimination on ChestMNIST

The DenseNet-121 reaches macro AUROC 0.729 (bootstrap 95% CI $[0.718, 0.738]$) on the 12,000-image test split, with macro AUPRC 0.135 and macro F1 0.031. The published MedMNIST v2 DenseNet-121 baseline for this dataset is approximately 0.77 [4]; ours is 0.04 AUROC below it. That baseline was trained on the full training split for many epochs on a GPU, while ours used $7{,}200$ images for four epochs on a CPU. Both statements are facts and we draw no inference from putting them side by side: a smaller budget explains nothing about whether the gap would close, and we did not test that.

What the per-label ordering does support is that the model learned thoracic structure rather than a dataset artifact (Fig. 3): readily visible findings rank highest (Edema 0.849, Cardiomegaly 0.830, Effusion 0.827) and subtle small ones lowest (Nodule 0.643, Infiltration 0.660), with every confidence interval clear of 0.5.

5.2 The operating point: silence on most of the taxonomy

AUROC is threshold-free, and it hides what this model does when asked for a decision. At the default 0.5 threshold, **11**

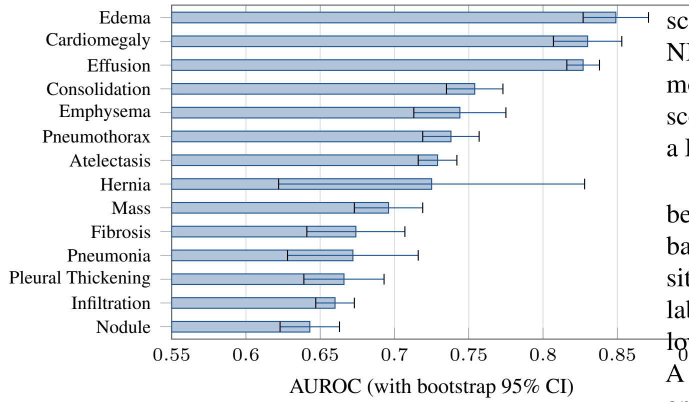


Figure 3: Per-label test AUROC on ChestMNIST (DenseNet-121, 64 px source upsampled to 224 px, CPU, best of four epochs, N_{test}=12,000), with bootstrap 95% CI whiskers, sorted ascending. The ordering tracks how visible each finding is, and every interval clears the 0.5 chance line.

Table 3: Operating point at the default 0.5 threshold (N = 12,000). Prevalence is $\text{support}_+/N$. Eleven labels (Atelectasis, Cardiomegaly, Mass, Nodule, Pneumonia, Consolidation, Edema, Emphysema, Fibrosis, Pleural Thickening, Hernia) emit no positive prediction at all, so their PPV is undefined.

Label	**Prev.**	**Sens.**	**Spec.**	**PPV**
Effusion	0.121	0.143	0.987	0.599
Infiltration	0.175	0.110	0.967	0.415
Pneumothorax	0.049	0.012	0.999	0.350
The other 11 labels	0.002–0.109	0.000	1.000	—
Macro (all 14)		0.019	0.997	0.097

of the 14 labels never produce a single positive prediction across all 12,000 test studies: their sensitivity is exactly 0.0, their specificity exactly 1.0, and their PPV undefined (Table 3). The three that fire at all are Effusion, Infiltration, and Pneumothorax, and Pneumothorax fires on 1.2% of its positive cases. Macro sensitivity is 0.019 and macro F1 is 0.031.

This is not a conservative operating point chosen for a clinical reason. Nothing was tuned: 0.5 is the default, and under prevalences from 17.5% (Infiltration) down to 0.2% (Hernia) a lightly trained model minimizes its loss by pushing almost every output below it. At this threshold the classifier is not usable as a detector. That is a property of the training budget and the destroyed resolution, not a design decision, and the AUROC of Section 5.1 should be read as a statement about *ranking* that carries no implication about decisions.

5.3 Calibration against a no-skill baseline

Read on their own, the calibration numbers look strong: macro Brier 0.045 and ECE 0.018. They are also close to meaningless here, and the arithmetic showing it needs no new experiment. For a label with prevalence p, a constant predictor that ignores the image and always returns p achieves a Brier score of exactly $p(1-p)$. Averaged over the 14 ChestMNIST prevalences, that is a macro Brier of $\mathbf{0.0472}$ for a model that has learned nothing at all. Our trained classifier scores $\mathbf{0.0453}$. The entire measured calibration advantage of a DenseNet-121 over a constant is 0.0019 Brier, about 4%.

We report this as a finding rather than an embarrassment, because it generalizes past our model. Under the class imbalance typical of multi-label radiology, where most labels sit below 5% prevalence, $p(1-p)$ is small for almost every label, so the no-skill floor of the aggregate metric is already low and there is little room between it and a perfect score. A model silent on 11 of 14 labels still posts a Brier of 0.045 and an ECE of 0.018, numbers that would pass unremarked in most results tables. Aggregate calibration metrics on imbalanced multi-label medical tasks should be reported against the constant-prevalence baseline, or not offered as evidence of quality at all.

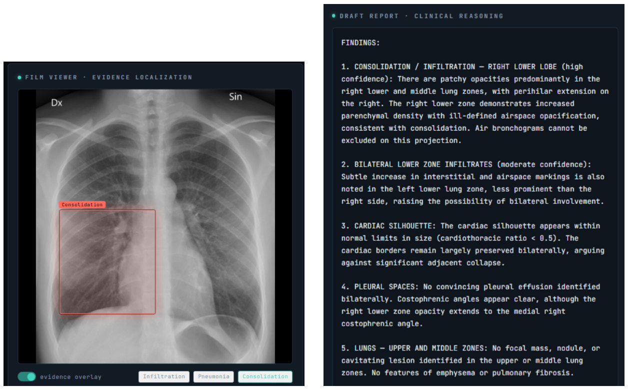
5.4 Harness control on synthetic data

To check that the training and scoring machinery credits only real signal, we train the same pipeline on a synthetic dataset whose generator injects a visible blob for seven of the fourteen labels and leaves the other seven with no visual correlate. A correct harness should learn the first group and sit near chance on the second, and it does: the seven signal-bearing labels reach a mean AUROC of 0.917 (Mass 0.979 down to Edema 0.866) and the seven no-signal labels average 0.533 (Hernia 0.620 down to Pleural Thickening 0.434). The no-signal mean sits slightly above 0.5 rather than on it, which is what a finite 350-image test split should give, and the individual no-signal labels straddle chance in both directions. The separation is what the control is for: the classifier, loss, metrics, and bootstrap intervals respond to injected signal and not to label frequency or ordering, so nothing is leaking.

5.5 Post-hoc invariance and latency

Layers 2 and 3 run after the forward pass and modify no weights and no logits, so the predictions are identical across the three conditions by construction. The harness confirms it: the maximum per-label probability change against the classification-only baseline is 0.000 over n=24 studies. This is a regression test, not a result. It earns its place because a nonzero delta would expose a real bug, an accidental coupling between the layers.

The cost that does vary is latency (Table 4). Prediction takes about 100 ms per study on CPU, the Grad-CAM pass adds 36 to 41 ms, and the offline template report adds 0.03 ms. The two conditions that run Grad-CAM measured 41.4 and 36.2 ms for the same work, which is the run-to-run spread of a 24-study CPU profile and is why the full-pipeline total (136 ms) came in below the localization-only total (141 ms). The test suite asserts the same invariance per modality against stubbed backbones, so the property is structural rather than



(a) Evidence localization

(b) Draft report

Figure 4: One study end to end on the **hosted serverless engine** (a vision LLM reading the image directly), not the benchmarked local stack. (b) is the worked example of Section 7: alongside its grounded findings the report states a cardiothoracic ratio and clear costophrenic angles, neither of which any component of the system measured.

particular to the chest checkpoint; no brain-MRI or head-CT measurement exists because no such checkpoint does.

6 Discussion

Finding-level grounding is a real guarantee and a narrow one. The useful thing MIRROR demonstrates is a boundary, and stating it precisely is worth more than the classifier behind it. Because the language layer receives a label, a probability, and a region name rather than an image, the set of findings the report may assert is exactly the set the classifier produced. That is checkable without trusting anything about the model: a reader holding the report and the probability vector can verify the correspondence, and a test can assert it. It is also strictly a claim about *which findings appear*, and says nothing about the sentences around them. A report that correctly asserts consolidation may frame it with clauses the system never computed, and those clauses inherit the credibility of the finding they sit beside (Section 7). This distinction is the thing worth taking from the system, and architectures that constrain only the former should say so rather than claiming groundedness generally.

Where this stands against Rudin. MIRROR is precisely the architecture Rudin [12] argues against for high-stakes decisions: an opaque CNN with a saliency map and a language layer attached after the fact. We have no rebuttal, and we do not claim the saliency map is faithful; the sanity checks of Adebayo et al. [11] are the right instrument for testing that and we have not run them. What we claim is narrower and does not depend on the heatmap being faithful at all: the finding set is auditable, because a reader can check the report against the probability vector directly. That is a retreat from explanation to auditability, and we would rather name it than dress it up. An inherently interpretable classifier would be the stronger answer.

On measuring the wrong thing. The ChestMNIST result is a systems demonstration: a real classifier, trained and scored end to end, whose per-label ordering tracks radiographic visibility and whose intervals clear chance. It is also unusable at its own default threshold, and the sequence of numbers it produces is instructive. AUROC 0.729 reads as a modest but working model. Brier 0.045 and ECE 0.018 read as a well-calibrated one. Both are true, and a model that emits no positive prediction for 11 of 14 labels sits behind them. Only the operating point exposes that, which argues for reporting sensitivity and PPV alongside every AUROC in this literature, and calibration against a prevalence baseline rather than on an absolute scale. Separately, serving one contract from both a PyTorch stack and a hosted vision LLM shows the *interface* MIRROR proposes is independent of the backend, at the price of the asymmetry in Section 3.5: the grounding guarantee belongs to the local stack, not to the deployed website.

7 Ethics, Safety, and Limitations

Not a medical device. MIRROR is a research prototype, not reviewed or cleared by any regulatory body; every output is a draft that must be verified by a licensed radiologist and must not be used for diagnosis or treatment. Each report carries that disclaimer explicitly.

A worked example of ungrounded detail. Fig. 4 shows a generated report beside its evidence overlay. Two phrases in it are worth quoting: the cardiac silhouette is described as normal *with a cardiothoracic ratio below* 0.5, and the *costophrenic angles are noted to be clear*. The system measured neither. No cardiothoracic ratio was computed, no cardiac or thoracic boundary was segmented, and no costophrenic angle was localized; the pipeline produced 14 probabilities and up to three saliency centroids, and nothing in that evidence supports either clause. Both are fluent radiology register generated to fill out a report. They also sit in the same paragraph as assertions that *are* grounded, and that adjacency is the danger: a reader who checks the consolidation finding against the overlay and finds it correct has been given a reason to extend trust to the sentence beside it. Fluency makes ungrounded detail more dangerous rather than less, because it arrives wearing the credibility of the verified finding next to it. This is why finding-level grounding cannot be described as making a report safe, and why the AI-generated-draft disclaimer is load-bearing rather than boilerplate.

Human subjects and data governance. No ethics-board review was required: this work uses only existing, public, de-identified benchmark datasets (NIH ChestX-ray14 and its MedMNIST derivative) and collects no new data from human subjects. Those datasets are governed by their own data use agreements; no raw images, weights, or protected

Table 4: Ablation conditions with measured per-stage latency (DenseNet-121 on ChestMNIST, profiled over 24 studies on CPU). **There is one AUROC measurement here, not three.** Macro AUROC $= 0.729$ is the single value from Section 5.1; it applies unchanged to every row because the post-hoc layers cannot alter a prediction, and the harness copies it into each row for that reason. The measurements that differ between rows are the latencies; the measurement verifying the rows are comparable is the probability delta.

Condition	**Classify**	**Localize**	**Report**	**Latency (ms/study)**
Classification only	✓			101 (prediction), total 101
+ Evidence localization	✓	✓		+41 (Grad-CAM), total 141
Full MIRROR	✓	✓	✓	+0.03 (template report), total 136

Macro AUROC $= 0.729$ for all three rows (one measurement, $N_{\text{test}} = 12{,}000$).
Maximum per-label probability change vs. the baseline $= 0.000$ over $n = 24$ studies.

health information are committed to version control, and the DICOM ingest extracts only non-PHI technical tags.

Limitations. Our one quantitative benchmark is ChestMNIST, at 64-pixel source resolution on a 7,200-image budget, and at its default threshold the classifier is not a working detector (Section 5.2). No clinical claim of any kind follows from it. Multi-modality is architectural: the brain-MRI and head-CT paths are implemented and exercised by the test suite but have no trained checkpoints, so MIRROR makes no predictive claim on those modalities. Explanation quality is unmeasured: the pointing-game and IoU protocol is implemented but not run, and no reader study has been conducted, so we have no evidence that the localization or report layers help a human, only that they cost about 40 ms. Class-activation maps are coarse and can highlight context rather than lesion [11], and we have not run the sanity checks that would detect it. The 3×3 location grid is a deliberately coarse descriptor. Report quality is assessed only qualitatively; a comparison against real radiologist reports (for example MIMIC-CXR [3]) is future work.

8 Conclusion

MIRROR composes classification, evidence localization, and report generation into one traceable, registry-driven pipeline, and the property worth carrying forward is a boundary rather than a score: constraining the language layer to structured evidence makes the *finding set* of a report auditable and does nothing for the prose around it. On ChestMNIST the classifier reaches macro AUROC 0.729 with a per-label ordering that tracks visibility, and at its default threshold it emits no positive prediction for 11 of its 14 labels while still posting a Brier score within 0.002 of a constant-prevalence predictor. We report that pair together deliberately: it is the clearest evidence we produced that aggregate metrics on imbalanced multi-label tasks can look healthy over a model that does nothing. Next steps are full-resolution training, the saliency sanity checks we cite but did not run, and a reader study, which is the only thing that would test whether any of this helps a radiologist.

Reproducibility

Code: `https://github.com/vignesh-nagarajan-vn/MIRROR`. Live demo (hosted engine): `https://mirror-ten-jet.vercel.app/`. A zero-setup CLI demo (`python -m demo.run_demo <image>`) runs the local pipeline on ImageNet weights with the offline template backend, no checkpoint or key required. Every results JSON stamps a `reproducibility` block (seed, git commit, library versions), and the unit tests are torch-free. The ChestMNIST result (`configs/chestmnist.yaml`) and the synthetic control regenerate from seed 42; outputs are committed under `results/`, and the no-skill baseline of Section 5.3 is recomputed from that JSON by `paper/calibration_baseline.py`.

Acknowledgments

This work was completed as part of the Global Indian Scientists & Technocrats (GIST) 2026 Summer Research Internship. The author thanks his mentor, **Sriram Venkatapathy**, an Applied AI Researcher at Capital One who holds a PhD in Computer Science from the Indian Institute of Technology (IIT) Hyderabad, for his guidance throughout the project: his experience building and evaluating machine-learning systems in industry shaped both the architecture and the emphasis on measurable, reproducible evaluation.

Declarations

Data availability: all datasets used are public, namely NIH ChestX-ray14 and its MedMNIST derivative ChestMNIST (CC BY 4.0); code, configurations, and result files are in the repository linked above. **Competing interests:** the author declares none. **Funding:** this work received no external research funding and was conducted as part of the GIST 2026 Summer Research Internship.

References

[1] X. Wang, Y. Peng, L. Lu, Z. Lu, M. Bagheri, and R. M. Summers. ChestX-ray8: Hospital-scale chest X-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. *CVPR*, 2097–2106, 2017.

[2] P. Rajpurkar, J. Irvin, K. Zhu, et al. CheXNet: Radiologist-level pneumonia detection on chest X-rays with deep learning. *arXiv:1711.05225*, 2017.

[3] A. E. W. Johnson, T. J. Pollard, S. J. Berkowitz, et al. MIMIC-CXR, a de-identified publicly available database of chest radiographs with free-text reports. *Scientific Data*, 6(1):317, 2019.

[4] J. Yang, R. Shi, D. Wei, Z. Liu, L. Zhao, B. Ke, H. Pfister, and B. Ni. MedMNIST v2: A large-scale lightweight benchmark for 2D and 3D biomedical image classification. *Scientific Data*, 10(1):41, 2023.

[5] G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger. Densely connected convolutional networks. *CVPR*, 4700–4708, 2017.

[6] M. Tan and Q. V. Le. EfficientNet: Rethinking model scaling for convolutional neural networks. *ICML*, 2019.

[7] A. Dosovitskiy, L. Beyer, A. Kolesnikov, et al. An image is worth 16x16 words: Transformers for image recognition at scale. *ICLR*, 2021.

[8] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra. Grad-CAM: Visual explanations from deep networks via gradient-based localization. *ICCV*, 618–626, 2017.

[9] H. Wang, Z. Wang, M. Du, et al. Score-CAM: Score-weighted visual explanations for convolutional neural networks. *CVPR Workshops*, 2020.

[10] J. Zhang, S. A. Bargal, Z. Lin, J. Brandt, X. Shen, and S. Sclaroff. Top-down neural attention by excitation backprop. *ECCV*, 2016.

[11] J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, and B. Kim. Sanity checks for saliency maps. *NeurIPS*, 2018.

[12] C. Rudin. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence*, 1(5):206–215, 2019.

[13] X. Wang, Y. Peng, L. Lu, Z. Lu, and R. M. Summers. TieNet: Text-image embedding network for common thorax disease classification and reporting in chest X-rays. *CVPR*, 2018.

[14] Z. Chen, Y. Song, T.-H. Chang, and X. Wan. Generating radiology reports via memory-driven transformer. *EMNLP*, 2020.

[15] G. Liu, T.-M. H. Hsu, M. McDermott, et al. Clinically accurate chest X-ray report generation. *MLHC*, 2019.

[16] E. Tjoa and C. Guan. A survey on explainable artificial intelligence (XAI): Toward medical XAI. *IEEE TNNLS*, 32(11):4793–4813, 2021.